

Notebook Manufacturing Scheduling

Mathematical Model

Mixed-Integer Programming (MIP) Formulation

Sets

Symbol	Name	Description
$\mathcal{I}$	Jobs	All production orders ($ \mathcal{I} = 198$)
$\mathcal{J}$	Machines	All machines in the factory ($ \mathcal{J} = 13$)
$\mathcal{R}$	Routes	All defined production routes ($ \mathcal{R} = 7$)
$\mathcal{T}$	Types	Product type groups ($ \mathcal{T} = 7$)

Route-machine compatibility: $ar(r, j) = 1$ if route r uses machine j . **Job-route compatibility:** $compat(i, r) = 1$ if job i can use route r .

Parameters

Symbol	Description
$p(i, r, j)$	Processing time of job i on machine j via route r (min/unit)
$s(i, i', j)$	Setup time on machine j when job i' follows job i (min)
$d(i)$	Due date of job i in minutes from today: $d(i) = \text{days} \times 1200$
$qty(i)$	Order quantity of job i (units)
$cap(j)$	Daily capacity of machine j (minutes)
$seq(r, j)$	Position of machine j in the sequence of route r
M	Big-M constant ($M = 9,999,999$)

Processing time rule:

$$p(i, r, j) = \begin{cases} p_{\text{cycle}}(i, r)/60 & \text{if } j \text{ is the bottleneck machine of route } r \\ p_{\text{cycle}}(i, r)/60 \times 0.25 & \text{otherwise (support operation)} \end{cases}$$

Setup time rule:

$$s(i, i', j) = \begin{cases} 40 \text{ min} & \text{if jobs } i, i' \text{ differ in ruling type and } j \in \{\text{Bielom1, Bielom2}\} \\ 0 & \text{otherwise} \end{cases}$$

Decision Variables

Variable	Domain	Meaning
$x(i, r)$	$\{0, 1\}$	1 if job i is assigned to route r
$z(i, i', j)$	$\{0, 1\}$	1 if job i is processed <i>before</i> job i' on machine j
$\delta(i)$	$\{0, 1\}$	1 if job i misses its due date
$y(i, j)$	$[0, 1]$	1 if job i is processed on machine j (derived via K2)
$C(i, j)$	≥ 0	Completion time of job i on machine j (min)
$C_i(i)$	≥ 0	Final completion time of job i (min)
$T_i(i)$	≥ 0	Tardiness of job i : $\max(0, C_i(i) - d(i))$ (min)
$C_{\max}$	≥ 0	Makespan: latest completion time across all jobs (min)

Objective Function

$$\min Z = \underbrace{\alpha \cdot C_{\max}}_{\text{makespan}} + \underbrace{\beta \cdot \sum_{i \in \mathcal{I}} T_i(i)}_{\text{total tardiness}} + \underbrace{\gamma \cdot \sum_{i \in \mathcal{I}} \delta(i)}_{\text{late job count}} \quad (1)$$

Implemented weights: $\alpha = 0.10$, $\beta = 0.60$, $\gamma = 0.30$, subject to $\alpha + \beta + \gamma = 1$.

Term	Weight	Purpose
$\alpha \cdot C_{\max}$	0.10	Minimize total production span; earlier batch completion
$\beta \cdot \sum T_i$	0.60	Penalise cumulative lateness; critical for priority orders
$\gamma \cdot \sum \delta_i$	0.30	Minimise number of customers affected by late delivery

Constraints

K1 — Unique Route Assignment

$$\sum_{r \in \mathcal{R}: \text{compat}(i,r)=1} x(i, r) = 1 \quad \forall i \in \mathcal{I}, \text{qty}(i) > 0 \quad (2)$$

Every job is assigned to *exactly one* compatible production route.

K2 — Route–Machine Consistency

$$y(i, j) = \sum_{\substack{r: \text{ar}(r,j)=1 \\ \text{compat}(i,r)=1}} x(i, r) \quad \forall i \in \mathcal{I}, j \in \mathcal{J} \quad (3)$$

Machine usage is fully determined by the route assignment; no independent choice.

K4 — Completion Time Lower Bound

$$C(i, j) \geq \sum_{\substack{r: \text{compat}(i,r)=1 \\ \text{ar}(r,j)=1}} p(i, r, j) \cdot \text{qty}(i) \cdot x(i, r) \quad \forall i \in \mathcal{I}, j \in \mathcal{J} \quad (4)$$

Completion time on a machine is at least the job's own processing time on that machine.

K5 — No-Overlap (Disjunctive) Constraints

$$C(i, j) \geq C(i', j) + \sum_r p(i, r, j) \cdot \text{qty}(i) \cdot x(i, r) + \sum_r s(i', i, j) - M(1 - z(i, i', j)) \quad \forall i \neq i', j \quad (5)$$

$$C(i', j) \geq C(i, j) + \sum_r p(i', r, j) \cdot \text{qty}(i') \cdot x(i', r) + \sum_r s(i, i', j) - M \cdot z(i, i', j) \quad \forall i \neq i', j \quad (6)$$

Exactly one ordering is active per pair. When $z(i, i', j) = 1$: job i completes before i' starts (Eq. 5 active). When $z(i, i', j) = 0$: job i' goes first (Eq. 6 active).

K6 — Job and Makespan Completion

$$C_i(i) \geq C(i, j) \quad \forall i \in \mathcal{I}, j : y(i, j) = 1 \quad (7)$$

$$C_{\max} \geq C_i(i) \quad \forall i \in \mathcal{I} \quad (8)$$

A job is done when it clears its *last* machine. Makespan is the latest over all jobs.

K7 — Tardiness Definition

$$T_i(i) \geq C_i(i) - d(i) \quad \forall i \in \mathcal{I} \quad (9)$$

$$T_i(i) \geq 0 \quad \forall i \in \mathcal{I} \quad (10)$$

$$\delta(i) \geq \frac{C_i(i) - d(i)}{M} \quad \forall i \in \mathcal{I} \quad (11)$$

T_i records minutes past deadline; $\delta(i)$ becomes 1 whenever any tardiness exists.

K8 — In-Route Machine Precedence

$$C(i, j') \geq C(i, j) + p(i, r, j) \cdot qty(i) \cdot x(i, r) \quad (12)$$

for all i, r, j, j' where $compat(i, r) = 1$, $ar(r, j) = ar(r, j') = 1$, and $seq(r, j') = seq(r, j) + 1$.

A job cannot start on the next machine in its route until it finishes on the current one.

K9 — Variable Domains

$$x(i, r), \delta(i), z(i, i', j) \in \{0, 1\}; \quad y(i, j), C(i, j), C_i(i), T_i(i), C_{\max} \geq 0 \quad (13)$$

Solver Configuration (CPLEX)

Parameter	Value
Solver	CPLEX (via GAMS)
Threads	4
MIP emphasis	1 (feasibility-first)
Optimality gap	10% (<code>epgap</code> = 0.10); final tolerance 5% (<code>optcr</code> = 0.05)
Time limit	3 600 s (CPLEX) / 7 200 s (GAMS)
$C_{\max}$ upper bound	2 500 000 min
Working day	1 200 min (20 hours); used to convert calendar days to minutes